Development of a Web-based Artificially Intelligent National Housing Cost Estimating System (NHCES) for Nigeria

THEMATIC AREA 24: Environment, Housing, & Urban and Regional Development

Executive Summary

Housing is a critical element of the 2030 Agenda for Sustainable Development, significantly influencing the achievement of multiple Sustainable Development Goals (SDGs). Actions toward adequate and affordable housing have multiple effects and support the implementation of other global development strategies, such as the New Urban Agenda, the Sendai Framework for Disaster Risk Reduction, and the Paris Agreement. These huge gains associated with adequate and affordable housing has made its provision a top priority, with developed and developing nations around the world all striving to close their respective housing deficits.

Nigeria, with a population exceeding 229 million, faces a staggering housing deficit estimated between 22 to 28 million units. According to the State House (2023), the funding requirement to meet Nigeria's housing needs is estimated at 21 trillion Naira. More recent reports put the estimate at 5.5 trillion annually over the next decade. The Nigerian government, as part of efforts to close its housing deficit, has created numerous housing policies and programmes backed up with robust financing initiatives such as the federal mortgage bank of Nigeria, Nigerian mortgage refinancing company, as well as public private partnership financing frameworks. However, no documented quick, objective, reliable and accurate means of estimating the cost of housing units to be constructed exist for Nigeria. As the funds form the financial initiatives are limited, the need to optimise the funds through early accurate estimates of the housing units cannot be overemphasized. This research aims to develop a Web-based Artificially Intelligent Housing Cost Estimating System that will aid quick, objective, reliable and accurate estimating of mass housing cost/budget for the government (MDAs) and other private clients in Nigeria.

Using a mixed research approach, this research will achieve its aim in two distinct phases. The first phase involves a descriptive study seeking to identify the existing cost estimation methodologies and the modifications necessary to develop a digital equivalent for the Nigerian context. The next phase handles development of the artificial intelligence component (using machine learning) of the system, and the entire development of the Web-based Artificially Intelligent National Housing Cost Estimating System. The expected short-term goal of this research is the provision of a cost estimating system that aid early cost information and advice to the government (MDAs) and other private clients for the purpose of realistic budgeting and optimization of the limited funds available for mass housing projects execution. The expected long-term goal on the other hand, is to provide the government of Nigeria with a useful basis for adequate cost planning and budgeting needed for successful implementation of its robust mass housing policies. This will aid the Nigerian government's determination to achieve many of the Sustainable Development Goals (SDGs) and also successfully align with the 2030 agenda for sustainable development. The research will be conducted in 24 months with an estimated budget of **N 42,500,004.75**.

Keywords: Housing, Cost estimation, Sustainable Development Goals, Nigeria, Artificial Intelligence

Duration of Research: _____24_____ months

1.0 General Research Background

1.1 Introduction

Housing is an important component of the 2030 agenda for Sustainable Development and an essential driver for achieving many of the Sustainable Development Goals (SDGs) as well as other global development strategies. Its provision offers several benefits including eradication of poverty for families, inequality reduction as well as building resilience against economic and natural disasters (Habitat for Humanity International, 2021).

Nigeria's determination to align with the 2030 agenda for Sustainable Development by providing adequate housing for its teeming population of around 229 million people, has been documented in numerous mass housing policies and programmes such as the National Housing Policies of 1991, 2002, 2006 as well as the Renewed Hope City Housing project lunched in 2023. In addition, robust financing initiatives such as the federal mortgage bank of Nigeria, Nigerian mortgage refinancing company, as well as public private partnership financing frameworks have been created to support these policies.

Nonetheless, despite these successive policies, reports indicate that as at June 2024, Nigeria still has a housing deficit of 22 to 28 million units. Furthermore, while it is projected that a total of 700,000 housing units are required yearly to close the deficit, not up to 10,000 housing units are being constructed annually due to bloated cost estimates/budgets.

Global best practice demands the use of national cost estimating systems such as the Building Cost Information Service (BCIS) of the UK and C5 cloud-based System of Australia to aid quick, reliable and accurate estimation of infrastructure project execution cost/budget. A fundamental benefit of these systems is provision of early cost advice to government and other clients which is key to realistic budgeting and fund optimising. They also provide a useful basis for benchmarking cost estimates of projects against proposed similar ones while adjusting for time and location differences. While massive successes have been reported with the use of such national cost estimating systems, they cannot be used in Nigeria since they are not built for the Nigerian macroeconomic environment.

Nigeria has only attempted to develop machine learning prediction models for construction material prices [1], determine macroeconomic indicators influencing construction material prices [2], modelled durations of construction projects [3, 4] and developed artificially intelligent e-procurement systems that could facilitate efficient procurement of construction projects [5]. These efforts are very limited and can't serve the purpose of an artificially intelligent cost estimating system.

Indeed, for Nigeria to achieve its massive aspirations in terms of mass housing development and the 2030 agenda for Sustainable Development, there is the need to develop an artificially intelligent national housing cost estimating system. This will provide Nigeria with a quick, reliable and accurate basis for estimating cost/budget needed for successful execution its national housing plans. It will also help to optimize funds available from the financial initiatives and even trigger planning on other means of raising funds. This research is aimed at developing an Artificially Intelligent National Housing Cost Estimating System (NHCES) for Nigeria, with a view to improving accuracy of mass housing cost/budget estimation.

1.2 Statement of the Problem

In a bid to ensure adequate housing, Nigeria has created numerous housing policies and programmes backed up with robust financing initiatives such as the federal mortgage bank of Nigeria, Nigerian mortgage refinancing company, as well as public private partnership financing frameworks. However, no documented quick, reliable and accurate means of estimating the cost of housing units to be constructed exist for Nigeria. As the funds form the financial initiatives are limited, the need to optimise the funds through early accurate estimates of the housing units cannot be overemphasised.

Although, globally, independent cost estimating systems such as Building Cost Information Service (BCIS) of the UK and C5 cloud-based software of Australia have been developed to facilitate early accurate cost estimates of infrastructure projects, their use cannot be generalised to all regions/countries as the underlying data as well as locational factors utilised for their development are specific to the economic regions/countries for which they were developed. In addition, they lack inbuilt machine learning capabilities that make a system intelligent and more versatile and robust. Clearly, there isn't a national housing cost estimating system with inbuilt machine learning capabilities for Nigeria. Nigeria must key into this global trend if it is to realise its national developmental aspirations which are so inevitably rooted in the massive provision of housing.

1.3 Aims, General and Specific Objectives of the Research Project

Aim & objectives

The main objective of this research is to develop a Web-based Artificially Intelligent National Housing Cost Estimating System (NHCES) for Nigeria, with a view to improving accuracy of mass housing project execution cost.

The specific objectives include:

1. Evaluate various methodologies of cost estimation of infrastructure projects and associated parameters.
2. Appraise the impact of macro-economic variables on cost estimates of housing projects.
3. Establish requirement models for the National Housing Cost Estimating System.
4. Develop conceptual models for the estimating system.
5. Develop appropriate machine learning models for estimating the cost of Housing.
6. Develop prototype web-based National Housing Cost Estimating System with inbuilt machine learning capabilities.

1.4 Conceptual Framework

The overall concept/principle of this study is the combination of housing cost data, key housing features influencing housing cost, the influence of macro-economic variables on housing cost as well as artificial intelligence, to develop a cost estimating system that provides a basis for quick, objective and reliable mass housing cost/budget as shown in figure 1.

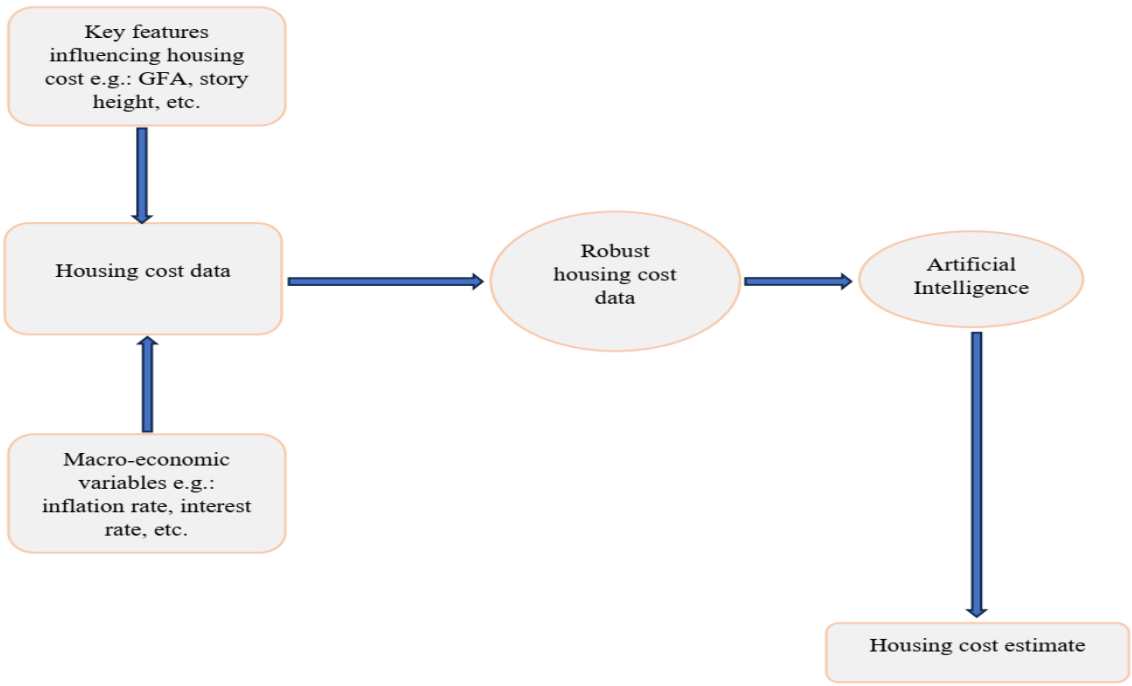

Figure 1: Conceptual framework

1.5 Project Goals

The immediate result of this research is a Web-based Artificially Intelligent Housing Cost Estimating System that will aid quick, reliable and accurate estimating of mass housing cost/budget for the government (MDAs) and other private clients in Nigeria. The expected short-term benefits include:

i. the provision of early cost information and advice to the government (MDAs) and other private clients for the purpose of realistic budgeting and optimization of the limited funds available for mass housing projects execution.
ii. provision of a useful basis for benchmarking cost estimates of projects against proposed similar ones while adjusting for time and location differences..

1.6 Project Impact

The project will provide the government of Nigeria with a useful basis for adequate cost planning and budgeting needed for successful implementation of its robust mass housing policies. It will boost the Nigerian government's opportunity to close its huge housing deficit and ensure that slum-like conditions that could lead to poor quality of life detrimental to health and education are completely eradicated. In addition, it will aid the Nigerian government's determination to achieve many of the Sustainable Development Goals (SDGs) and also successfully align with the 2030 agenda for sustainable development.

2.1 Literature Review

Traditional cost estimating approach

The traditional cost estimation approach has dominated the construction domain for several decades. The approach primarily depends on expert judgment, parametric estimation, analogous analysis, and bottom-up estimation (Aram et al., 2014; Barakchi et al., 2017; Juszczyk, 2017; Molcho et al., 2014). Expert judgment relies on the insights and experiences of seasoned estimators who draw from their involvement in similar past projects. In contrast, parametric estimation employs mathematical models that establish a correlation between costs and relevant cost parameters derived from previously completed projects. Analogous analysis, on the other hand, estimates costs in the early stages of a project by comparing a specific criterion's value from a prior project, such as duration, to that of the new project. Bottom-up estimation, however, focuses on calculating the actual costs associated with each activity within the project, encompassing both direct costs, such as labour and materials, and indirect costs, such as overhead expenses from the head office.

To improve the traditional approach to cost estimation, numerous studies have been conducted to identify the key factors influencing the construction costs of housing units. For example, Das et al. (2009) estimated the annual growth rate of housing unit prices in South Africa, taking into account various factors such as construction costs, stock market fluctuations, income levels, and interest rates. Additionally, Myers (2016) conducted research focused on predicting construction costs by examining the expenses associated with construction materials, labour, and equipment. Fereshtehnejad and Shafieezadeh (2016) explored construction costs through the lens of multiple variables, including the use of recycled materials, project duration, type, and scheduling. Furthermore, economic indicators such as the wholesale price index (WPI), liquidity, and building services index (BSI) have been found to significantly influence construction costs, particularly in times of economic uncertainty (Adeli and Karim, 1997; Karim and Adeli, 1999; Hsiao et al., 2012; Jafarzadeh et al., 2014; Rafiei and Adeli, 2018). Additionally, various computational methods have been applied in the construction sector to address a range of challenges, including construction simulation (Yu et al., 2017), labour assignment (Yi and Wang, 2016; 2017), and team allocation (Florez, 2017).

Even with these efforts, the traditional estimation methods exhibit several limitations, primarily due to their dependence on the estimator's educational background and experience, which can result in subjective and error-prone decisions (Ekung et al., 2021). Moreover, these conventional techniques are often labour-intensive and hindered by a shortage of qualified personnel possessing adequate construction expertise. Consequently, there is a growing imperative to explore alternative techniques that can effectively replace traditional methods.

Artificial intelligence-based cost estimating approach

Numerous studies have focused on using artificial intelligence to enhance the overall efficiency of housing cost estimates. For example, Wang and Ashuri (2017) explore the utilisation of random forests to enhance the precision of the construction cost index, which reflects the overall cost dynamics within the United States market. This index plays a critical role in influencing the accuracy of project cost estimations, leading to potential underestimations or overestimations. Their findings indicate that the tree ensemble method outperforms both neural networks and conventional estimation techniques in terms of predictive accuracy. More recently, Habib et al. (2025) present a decision-support framework based on ensemble learning, which integrates regression random forests and gradient-boosting regression trees via regression voting to automate the cost estimation process for both residential and commercial projects.

Robust cost databases

Globally, attempts improve on the traditional cost estimating methods and ensure accurate estimation of project execution costs/budgets have advanced to the use of national cost databases. For instance, the Royal Institution of Chartered Surveyors (RICS) has developed the Building Cost Information Service (BCIS) which collects cost and price information across the UK construction industry and then collates, analyses, models, interprets and makes it available to the industry to facilitate accurate cost planning and modelling. Australia developed the C5 cloud-based estimating software for project organisation, pricing, distribution of markup in a variety of ways, analysis and customisation of tender submission, and also clear reporting of bids for internal or external (Client) purposes. China developed a flexible spatiotemporal model (FSTM) capable of exploring the spatiotemporal characteristics of residential house price and the impact factors in middle-small cities. The model integrates both spatial and temporal components, accounts for spatiotemporal characteristics and reproduce the spatial variability and temporal trends of residential house price in China (Wang et al., 2022).

Among the developing countries, Iran developed a housing price model based on the gene expression programming (Shekarian and Fallahpour, 2013), while Malaysia, developed the Building Cost Information Services Malaysia (BCISM) for the construction industry in Malaysia. The key functionalities of these systems are to: provide advisory on construction economics to the government for policy making, assist in preparing construction budget estimates, cost plans, cost benchmarks, costing forecast, Building Information Modelling (BIM), etc.

This study seeks to overcome the shortcomings associated with the traditional approach to cost estimation for housing systems in Nigeria. The study will focus on areas such as robust data gathering, artificial intelligence and the reduction of subjective biases. Furthermore, the study will incorporate economic considerations, project-specific factors, and real-time data to facilitate more adaptive and accurate cost estimations in mass housing.

2.2 Research Methodology

This study will achieve its aim in two distinct phases. The first phase involves a descriptive study seeking to identify the existing cost estimation methodologies and the modifications necessary to develop a digital equivalent for the Nigerian context. The next phase handles the development of the artificial intelligence component (using machine learning) of the system, and the entire development of the National Housing Cost Estimating System.

The first phase of the methodology applies to objectives 1 to 4 of this study. Specifically, a mixed research approach involving a combination of survey and archival analysis research styles will be applied to paint a comprehensive picture of the functional requirements for the National Housing Cost Estimating System. An extensive review of literature will be conducted to identify the existing cost estimation methodologies and associated parameters, while interviews and questionnaire survey will be conducted to validate and facilitate necessary modifications to the identified methodologies and associated parameters. The archival analysis on the other hand, will review national and local databases in both manual and digital records to collect data relevant to the identified parameters. Some of the databases that will be consulted include: the National Bureau of Statistics, Central Bank of Nigeria and other reports from relevant agencies. Residential housing completed within the last 10 years will form the study's population. The sample size for the study will be calculated using Cochran's table while purposive sampling will be used to elicit responses from construction firms and real estate developers. Data obtained from the survey and archival analysis will be analysed using descriptive statistics (Mean score, standard deviation, normalised mean) and inferential statistics (T-test, ANOVA, factor analysis).

The second phase of the methodology addresses objectives 5 and 6 of the study. The RIPPLE and Unified Process methodologies, which are object-oriented methodologies will be used to guide the development of the system. The methodologies operate on the spiral, iterative and incremental methodologies of the Software Development Lifecycle. The input for the RIPPLE methodology is the output of the first phase above that has been achieved using the quantitative research approach i.e. business requirements modelling (BRM). This describes the working system of the current/existing cost estimation process with the aid of numerous UML artefacts including use cases, actor lists, activity diagrams, and communication diagrams. The complete system development methodology for the Intelligent National Housing Cost Estimating System (iNAHCES) is discussed in detail below and summarised in Figure 1.1.

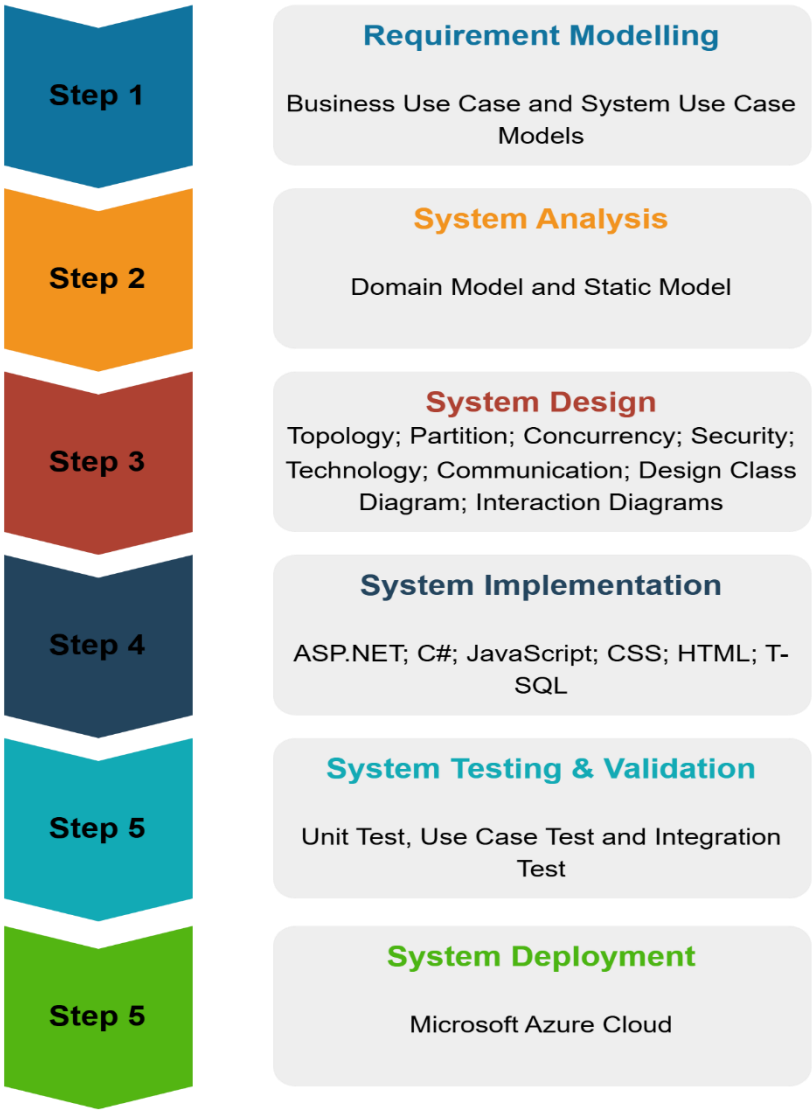

Figure 1.1. Software Development Methodology for iNAHCES

Step 1: Requirements Gathering

This is the stage where the requirements for the proposed iNAHCES are captured. The requirements are established from the findings of objectives 1 and 2. This fulfils the requirements gathering step of the RIPPLE methodology which encompasses the development of a Business Requirement Model (BRM). The BRM illustrates how the manual housing cost

estimation process works. A System Requirement Model (SRM) which describes the specified functionalities of the proposed iNAHCES is developed based on the identified requirements from the BRM. The BRM and the SRM will be depicted using several UML artefacts i.e. use cases, actor lists, use case details, use case surveys, activity diagrams, communication diagrams, interface sketches, and glossary. This stage will be achieved in line with the established requirements gathering steps in previous studies (Yamusa et al., 2023).

Step 2: System Analysis

The study will develop the domain model and static analysis model under this stage. The domain model falls under the Unified Process (UP) methodology while the static model falls under the RIPPLE methodology. The function of the domain model generally lies in helping to understand the existing housing cost estimation process before the commencement of the development of an equivalent digital version. On the other hand, the static model involves deciding on the logical or physical parts of the iNAHCES system and how they should be interconnected. It also describes how to construct and initialise the proposed iNAHCES. The domain model will be developed primarily using the BRM in Step 1. The static model will be developed primarily based on the SRM in Step 1. Both models will then serve as inputs into the operation contracts and design model of the UP methodology.

Step 3: System Design

The system design will employ a similar approach as previous studies (Abdullahi et al., 2019; Abdullahi et al., 2022; Yamusa et al., 2024). The system design deals with issues such as system topology, technology choices, concurrency choices, security policy, subsystem partition, partitioning subsystem into layers, interface design, and deciding how the machine, subsystem and layers can communicate. In addition, using the UP approach, system sequence diagram, operation contracts, design class diagrams and interaction diagrams will be developed for iNAHCES at this stage and used as the basis for the implementation of the system.

Step 4: System Specification and Implementation

The coding for the iNAHCES will be carried out using Microsoft ASP.NET technology and C# programming language within the Microsoft Visual Studio integrated development environment. ASP.NET is a web development platform supported within the .NET framework for the creation of dynamic web applications. Other technologies that will be used in the development of iNAHCES are HTML, CSS, JavaScript, and Microsoft SQL Server. These tools have proved useful in the development of previous systems that have been tested and validated (Abdullahi et al., 2019; Abdullahi et al., 2022; Yamusa et al., 2024). For Machine Learning (ML), the Microsoft Azure Machine Learning tool plus Python will be utilised to develop the various machine learning models (as in Abdullahi et al., 2024a,b; Yamusa et al., 2024). The ML models will be developed using the steps involved in previous studies (Abdullahi et al., 2024a,b; Ahmadu et al., 2023). The steps are summarised in Figure 1.2. This study will employ the SHapley Additive exPlanation (SHAP) to interpret the contributions of each variable and factor to the predictive model.

Step 5: System Testing, Validation, and Deployment

The iNAHCES will be tested within the Visual Studio environment using the debugging tools provided. The system will be tested for various types of errors i.e. syntax, logical and runtime errors. The iNAHCES will be validated by procurement professionals with deep experience in the procurement process to ensure that it correctly represents the initial problem domain model. The developed system will be deployed to Microsoft Azure as Software as a Service (SaaS) model and will be merged with the ML models that will be developed using Microsoft Azure ML tool.

Validation of the Intelligent National Housing Cost Estimating System

The developed Intelligent National Housing Cost Estimating System needs to be validated by experts to ensure that it represents the initial problem domain i.e. manual housing cost estimation process. There are many ways of carrying out analytical model validation of the

computerised system. The model validation process proposed by Martis (2006) and as implemented by AbdulAzeez, (2012) will be adopted for this research. The Martis's model validation process involves three types of validations i.e. conceptual model validation, computerised model validation, and operational validation. In addition, a Business Process Reengineering (BPR) tool will be used to assess the performance of the iNAHCES compared to the existing manual-based system. Concerning the integrated ML models in iNAHCES, the predictive power of the models will be assessed by comparing their performances in new data stream mode versus training mode.

Figure 1.2. Methodology for Building Machine Learning Models.

2.3 Research Activity/Output indicators

The major activities of the project and the output expected at the end of each activity is summarised in Table 1.1.

Table 1.1: Major activities of the project and the output expected

STEPS/METHODS	RESEARCH PROGRAMME	OUTCOMES
Step 1: Literature Review	-Review of various methodologies of cost estimation with reference to their parameters -Review of macro-economic factors influencing cost estimates of housing	- Identified methodologies of cost estimation with reference to their parameters - Identified macro-economic factors influencing cost estimates of housing

	construction projects in Nigeria -Review of variables and indexes influencing cost estimates of housing construction projects in Nigeria	construction projects in Nigeria - Identified variables and indexes influencing cost estimates of housing construction projects in Nigeria
Step 2: Field Study	- Questionnaire distribution; Interviews; and collection of archived cost data on housing. - Presentation, analysis and discussion of data	- Assessed/appraised parameters for cost estimating - Assessed/appraised macro-economic factors influencing cost estimates of housing construction projects in Nigeria - Assessed/appraised variables and indexes for cost estimates of housing construction projects in Nigeria. - Actual cost data on housing projects
Step 3: Requirements Modelling	-Development of Business Use-Case Models and System Use Case Models using UP & UML	-Business Use-Case Models -System Use-Case Models
Step 4: System Analysis, Design, Implementation, Testing, Validation, and Deployment	- Development of machine learning models to support forecast using Ripple and Unified Process methodologies; UML; ASP.NET + C#, python - System Validation	-Validated machine learning models to support cost estimating

Step 5: Summary, Conclusion and Recommendations	- Write summary, conclusion and recommendations	Answer the primary research questions

2.4 Time Frame: [Provide a timeline for the major activities of the project.]

S/N	DESCRIPTION OF ACTIVITIES	DURATION	YEAR	QUARTER			
				1st	2nd	3rd	4th
Stage 1: Data Gathering							
1	Extensive review of literature to identify existing cost estimating methodologies and associated parameters	3 Months	2025				
2	Survey and archival analysis (for validation and collection of relevant data)	6 Months	2025				
Stage 2: System Development							
3	Statistical Analysis: Preliminary analysis of data	3 Months	2026				
4	Development of web-based National Housing Cost Estimating System with inbuilt machine learning capabilities	7 Months	2026				
5	Conclusion	2 Months	2026				
Stage 3: Research Dissemination							
6	Writing and Publishing	3 Months	2026				

24 Months

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2.5 Key Performance Indicators

MAJOR ACTIVITIES		INDICATORS
Step 1: Literature Review	-Review of various methodologies of cost estimation with reference to their parameters -Review of macro-economic factors influencing cost estimates of housing construction projects in Nigeria -Review of variables and indexes influencing cost estimates of housing construction projects in Nigeria	- Number of research papers reviewed - Quality/credibilty of literature sources - Relevance of literature review outcome to the research work - Completion within stated timeline.
Step 2: Field Study	- Questionnaire distribution; Interviews; and collection of archived cost data on housing. - Presentation, analysis and discussion of data	- Completion within stated timeline - Response rate/percentage - Quality and number of archival records reviewed
Step 3: Requirements Modelling	-Development of Business Use-Case Models and System Use Case Models using UP & UML	-Completion within stated timeline
Step 4: System Analysis, Design, Implementation,	- Development of machine learning models to support forecast using Ripple and Unified Process methodologies; UML; ASP.NET + C#, python - System Validation	- Completion within stated timeline - Accuracy of the developed model/system

Testing, Validation, and Deployment		
Step 5: Summary, Conclusion and Recommendations	- Write summary, conclusion and recommendations	- Number of publications from the research work

2.6 Monitoring and Evaluation Mechanism

Monitoring and Evaluation will be on the basis of three key performance indicators namely, quality, time, and cost.
Quality Performance: The quality of the research outputs will be monitored and evaluated on the basis of the "Activities and Outcomes" outlined in section 2.3 and the associated activity indicators in section 2.5.
Time Performance: The time performance of the project will be monitored and evaluated on the basis of "Time Frames" outlined in section 2.4.
Cost Performance: The cost performance of the project will be monitored and evaluated on the basis of the "Project Budget" in section 4.1.

Generally, the PI shall draft a Monitoring & Evaluation (M&E) plan where project members are assigned specific roles and responsibilities and associated quality, time and cost targets. The PI shall be responsible for the implementation of the M&E plan by collecting individual project members' progress reports and analysing them to determine the actual level of performance achieved to the projected performance expected. Where corrective measures are required, the PI shall ensure that actions are taken by responsible parties to restore the project to the expected course.

Coordination Under the M&E Structure

The Ahmadu Bello University has a general structure for coordinating such research activities through the Directorate of Academic Planning and Monitoring (DAPM) Unit. In addition to the M&E functions of the Principal Investigator, the DAPM unit shall serve as an additional M&E layer to ensure tighter monitoring and control of project progress.

2.7 Dissemination Strategies

The steps to be taken are:

1. Publications of research papers in SCI-indexed Journals.
2. Attending Local and International Conferences to present research findings.

3. Demonstration of the prototype national housing cost estimating system (through workshops and seminars) to critical stakeholders e.g. ABU, NUC, Nigerian Institute of Quantity Surveyors (NIQS) etc.

3.1 Composition of the Research Team

A. Principal Researcher

Name and Position of the Principal Researcher: Hassan Adaviriku Ahmadu
Date of Birth: 15th June, 1984 Gender: Male E-mail Address: haahmadu@abu.edu.ng, ahmaduhassan@rocketmail.com
Highest Qualification and Area of Specialization: PhD Quantity Surveying / Modelling in Construction
Rank: Reader
Name of Organization: Ahmadu Bello University, Zaria
Faculty/Department: Environmental Design/Quantity Surveying
Postal Address: Department of Quantity Surveying, Faculty of Environmental Design, Ahmadu Bello University
City/Town: Zaria Telephone Number (Mobile): 08055419494, 08136516216 Fax:

B.1 Research Partner:

Name and Position of the Researcher: Yahaya Makarfi Ibrahim
Date of Birth: 15/02/1974 Gender: Male E-mail Address: ymibrahim@abu.edu.ng, makarfi@gmail.com
Highest Qualification and Area of Specialization: Professor of Quantity Surveying / Construction Project Management & Software
Rank: Professor
Name of Organization: Ahmadu Bello University, Zaria
Faculty/Department: Environmental Design / Quantity Surveying
Postal Address: Department of Quantity Surveying, Faculty of Environmental Design, Ahmadu Bello University

B.2 Research Partner:

Name and Position of the Researcher: Joy Joshua Maina
Date of Birth: 15/02/74 Gender: Female E-mail Address: jjmaina@abu.edu.ng
Highest Qualification and Area of Specialization: Professor of Architecture / Housing, Spatial Analyses and STEM Education
Rank: Professor
Name of Organization: Ahmadu Bello University, Zaria
Faculty/Department: Environmental Design / Architecture
Postal Address: Department of Architecture, Faculty of Environmental Design, Ahmadu Bello University

B.3 Research Partner:

Name and Position of the Researcher: Bello Abdullahi
Date of Birth: 12/02/1976 Gender: Male E-mail Address: abdullbello@yahoo.com; baeconsultingengineers@gmail.com
Highest Qualification and Area of Specialization: PhD in Construction Management / Application of IT in Construction
Rank: Principal Partner / Industry Expert
Name of Organization: BAE Consulting Engineers
Faculty/Department: N/A
Postal Address: Flat 3 Block 4 Kaunda Close OAU Quarters off Yusuf Maitama Street Asokoro, Abuja
City/Town: Abuja Telephone Number (Mobile): 08060357365 Fax:

B.4 Research Partner:

Name and Position of the Researcher: Muhammad Aliyu Yamusa
Date of Birth: 10/10/1994 Gender: Male E-mail Address: yamusajf@yahoo.com
Highest Qualification and Area of Specialization: MSc / Construction Informatics; Machine Learning; Natural Language Processing; Web based Systems
Rank: Lecturer II
Name of Organization: Ahmadu Bello University, Zaria
Faculty/Department: Environmental Design / Quantity Surveying
Postal Address: Department of Quantity Surveying, Faculty of Environmental Design, Ahmadu Bello University

C.1 Research Mentees / Young Academics:

Name and Position of the Researcher: Musa Ayuba Musa
Date of Birth:07/10/1990 Gender:Male E-mail Address:musaayubamusa@gmail.com

Highest Qualification and Area of Specialization: MSc Quantity Surveying/ ICT in Construction
Rank: Lecturer II
Name of Organization: Baze University, Abuja
Faculty/Department: Environmental Sciences / Quantity Surveying
Postal Address: Department of Quantity Surveying, Faculty of Environmental Sciences, Baze University, Abuja

C.2 Research Mentees / Young Academics:
Name and Position of the Researcher:
Date of Birth:00/00/19 Gender:Male E-mail Address:
Highest Qualification and Area of Specialization: /
Rank:
Name of Organization: Ahmadu Bello University, Zaria
Faculty/Department: Environmental Design /
Postal Address: Department of ….., Faculty of Environmental Design, Ahmadu Bello University

3.2 Profile

3.3 Research Work to Date

i. Ahmadu, H. A., Ibrahim, Y. M., Abdulrahman, R. S., Jibril, U. S. and Yamusa, M. A. (2023). Developing Machine Learning Prediction Models for Construction Material Prices in Nigeria. *Journal of Environmental Technology, A.T.B.U. Bauchi, Nigeria,* 16(1), pp.22-42
ii. Ahmadu, H. A., Ibrahim, Y. M., Abdulrahman, R. S., Jibril, U. S., & Yamusa, M. A., **(**2023). "Assessing Macroeconomic Indicators Influencing Construction Material Prices in Nigeria". *ENVIRON Journal of Environmental Studies,* 5(1), pp. 163-174
iii. Ahmadu, H. A., Ibrahim, A. D., Ibrahim, Y. M., and Adogbo, K. J. (2020). Incorporating aleatory and epistemic uncertainties in the modelling of construction duration. https://doi.org/10.1108/ECAM-06-2019-0304
iv. Ahmadu, H. A., Ibrahim, Y. M., Ibrahim, A. D., and Abdullahi, M. (2015). Modelling Building Construction Durations. https://doi.org/10.1108/JFMPC-02-2014-0004
v. Yamusa, M. A., Ibrahim, Y. M., Abdullahi, M., Ahmadu, H. A., Abdullahi, B., Ibrahim, A. D., and Bala, K. (2024). Developing a Machine Learning Integrated E-Procurement System for Nigerian Public Procuring Entities. http://dx.doi.org/10.1504/IJPM.2023.10055600.
vi. Maina, J. J. (2023). Factors influencing Transformation of Prototype Public Housing in Northern Nigeria. *Covenant Journal of Research in the Built Environment, 11*(1), 46-67. https://journals.covenantuniversity.edu.ng/index.php/cjrbe/article/view/3930.
vii. Yamusa, M. A., Ibrahim, Y. M., Abdullahi, M., Ahmadu, H. A., Abdullahi, B., Ibrahim, A. D., Bala, K., & Aminu, A., 2023. "Developing Requirement Model for a Web-Based E-Procurement System in Nigeria". International Journal of Electronic Governance, 15(3), pp. 234-254. http://dx.doi.org/10.1504/IJEG.2023.133573.
viii. Abdullahi B., Ibrahim Y. M, Ibrahim A. D. and Bala K. (2022). Development of web-based e-Tendering system for Nigerian public procuring entities, International Journal of Construction Management, published by Taylor & Francis 22(2), 278-291, DOI: 10.1080/15623599.2019.1620492
ix. Abdullahi, B., Ibrahim, Y.M., Ibrahim, A.D., Bala, K., Ibrahim, Y., & Yamusa, M. A. (2024a). "Development of Machine Learning Models for Classification of Tenders based on UNSPSC Standard Procurement Taxonomy". *International Journal of Procurement*

Management, Inderscience, 19(1), pp. 445 – 472, http://dx.doi.org/10.1504/IJPM.2023.10056755

x. Abdullahi, B., Ibrahim, Y.M., Ibrahim, A.D., Bala, K., Ibrahim, Y., & Yamusa, M. A. (2024b). "Development of Machine Learning Models for Categorisation of Nigerian Government's Procurement Spending to UNSPSC Procurement Taxonomy". *International Journal of Procurement Management*, Inderscience, 19(1), pp. 106 – 121, https://doi.org/10.1504/IJPM.2024.135175

xi. Musa, M. A., Abdullahi, M., Ahmadu, H. A., Ibrahim, A. M., & Ibrahim, A. D. 2024. "Framework for design bid and build automated construction cost management process in Nigeria". In: Laryea, S., Li, B., Essah, E. A., Mensah, S., Liu, H., and Yao, R. (Eds). Proceedings of the WABER & SuDBE Joint International Conference, Wits University Johannesburg, South Africa, 29th July – 2nd August, 2024, pp. 11-19.

xii. Musa, M. A., Abdullahi, M., Ahmadu, H. A., Ibrahim, A. M., & Ibrahim, A. D. 2024. "Integrating construction cost management knowledge in large language models through prompt engineering". Proceedings of 1st International Conference on Procurement, Environment and Social Standards: Key to a Sustainable Livelihood in the 21st Century, Federal University of Technology Owerri, Nigeria, 11th – 15th November, 2024

3.4 Previous Research Grant

i. TETFund IBR Grant, *"Developing Machine Learning Models for Construction Material Prices in Nigeria"*. - N1,233,500.00 million; 2022 Institution Based Research Grant, Tertiary Education Trust Fund (TETFUND), Batch 7 (TETF/DR&D/UNI/ZARIA/IBR/2020/VOL.1/22). Duration: 12 months, Status: Completed.

ii. National Research Fund, Tertiary Education Trust Fund (TETFUND) 2019. "Development of e-Procurement System for Nigerian Public Procuring Entities". N31,150,000.00 million. Duration: 24 months, Status: Completed.

iii. TETFund IBR Grant, *An Evaluation of the Near-campus Housing Environment of Samaru and its relationship with the Quality of Life of Residents and Academic Performance of Students* - N3,000,000; TETF/DR&D/UNI/ZARIA/IBR/2024/Batch 8/04; Oct. 2024-September 2025. Duration: 12 months, Status: Ongoing.

3.5 Group Research

Research team member	Role	Responsibilities
Principal Investigator	Lead the research project and responsible for its overall success	Coordinate the research team and ensure milestones are met.
		Oversee data collection, analysis, and reporting
		Web-based Artificially Intelligent Cost Estimating System development
		Manage the grant budget
Co-investigators	Work closely with PI and share	Research design and methodology
		Assist in data collection, analysis and interpretation

	responsibilities for project execution	Web-based Artificially Intelligent Cost Estimating System development
Research assistants	Support the PI and co-investigators in various research activities	Conduct literature reviews and background research
		Collect and analyse data under the guidance of PI and co-investigators
		Assist in writing research reports, papers, and presentations under the guidance of PI and co-investigators

4.1 Project Budget

SN	DESCRIPTION OF ITEM	EXPECTED FROM		TOTAL
		TETFund NRF	INSTITUTION	
1.0	**Personnel Costs/Allowances**			
1.1	Principal Researcher 20 man-month @ 100,000 per man-month	**2,000,000.00**		
1.2	Team Members: 4 researchers for 80 man-months @ 70,000 per man-month	**5,600,000.00**		
1.3	Technical Support: 1 front/back-end developer for 20 man-months @ 45,000 per man-month	**900,000.00**		
1.4	Others (Please specify)			
	Sub-Total (Not >20% of budget)	**8,500,000.00**		
2.0	**2.0 Equipment (List & Specify)**			
2.1	4 Nos High Performance laptop for researchers: Apple Macbook 15.4" Retina Quad-Core i7 2.8Ghz 16GB 256GB with touch ID and Touch Bar	**5,500,000.00**		
2.2	1 Nos Desktop for research technical assistants: HP Pavilion All-in-One Intel core i5 12 GB RAM, 1 TB HDD	**1,200,000.00**		
2.3	5 Nos Officejet Pro 9013 All-in-One Printer	**870,540.00**		
2.4	24 months Subscription of Cloud-based Microsoft Visual Studio Professional for NHCES software development	**3,054,460.00**		
	Sub-Total (Not > 25% of budget)	**10,625,000.00**		
3.0	**3.0 Supplies/Consumables**			
3.1	Printer Ink	**350,000.00**		
3.2	Stationaries	**600,000.00**		
3.3	Internet Subscription	**1,500,000.00**		
	Sub-Total	**2,450,000.00**		
4.0	**4.0 Data Collection & Analysis**			
4.1	Research Assistants	**2,550,000.00**		
4.2	Research Informants	**2,550,000.00**		
4.3	Technical Assistants	**2,550,000.00**		
4.4	Collection Instruments	**350,000.00**		

4.5	Data Analysis	**850,000.00**			
	Sub-Total	**8,850,000.00**			
5.0	**5.0 Travels**				
5.1	Local travels to Lagos, Port-Harcourt, Enugu, Abuja. Kano, Jos, Borno to collect research data covering flight ticket, local transport, accommodation, feeding	**5,151,195.00**			
	Sub-Total	**5,151,195.00**			
6.0	**6.0 Dissemination**				
6.1	Demonstration of system to key stakeholders within Nigeria expenses	**1,600,000.00**			
	Sub-Total (Not >3%)	**1,600,000.00**			
7.0	**7.0 Others/Miscellaneous (Specify)**				
7.1	Communications	**1,000,000.00**			
7.2	Refreshment for meetings	**1,300,000.00**			
7.3	Other unforeseen expenses	**1,000,000.00**			
	Sub-Total	**3,300,000.00**			
	TOTAL DIRECT COST	**40,476,195.00**			
	INDIRECT COST (5% of TETFund Component of Direct Cost) to Institution	**2,023,809.75**			
	GRAND TOTAL	**₦ 42,500,004.75**			

4.2 Budget Justification

The research budget is driven by the major activities required to achieve the research objectives and the overall aim. Specifically, the questionnaire survey, interviews and archival analysis will require a lot of local travels to Lagos, Port-Harcourt, Enugu, Abuja, Kano, and Gombe to collect data and also interact with key stakeholders during the data collection and validation stages. Furthermore, the development of the web-based artificially intelligent estimating system would require the input of technical experts and high-capacity computers and subscriptions to web domains and other cloud facilities. Additionally, dissemination of the results will require attending local and international conferences and workshops which may require a lot of travelling and use of internet.